Google Gemini Model Selection

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Overview

Overview of Gemini Models

Data Flow for Text Generation

Adjusting Model Interactions

Gemini Overview

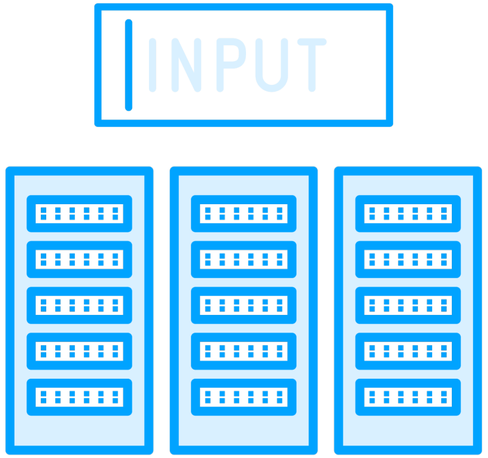
Based on Transformers

Contain a large number of neural layers

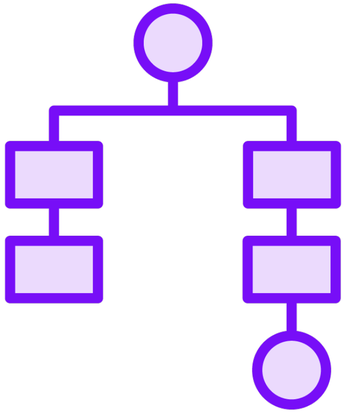
Slow and Costly

We should choose appropriate model interactions for each use case

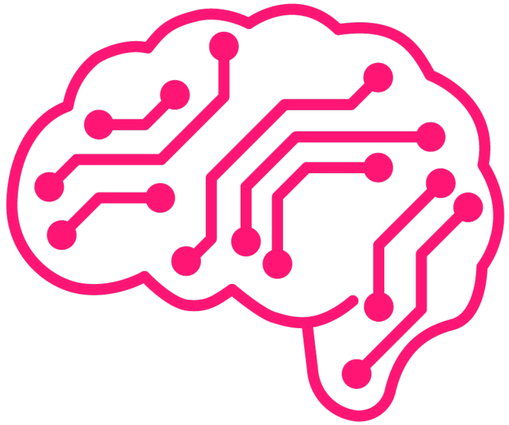
Data Flow for Generation



Send input data to Gemini



Convert to final input data based on prompt structure



Model generates output based on base model and parameters

Base Models

Based on Size:

Speed vs Quality

Based on Stability:

Quality vs Predictability

Model Parameters

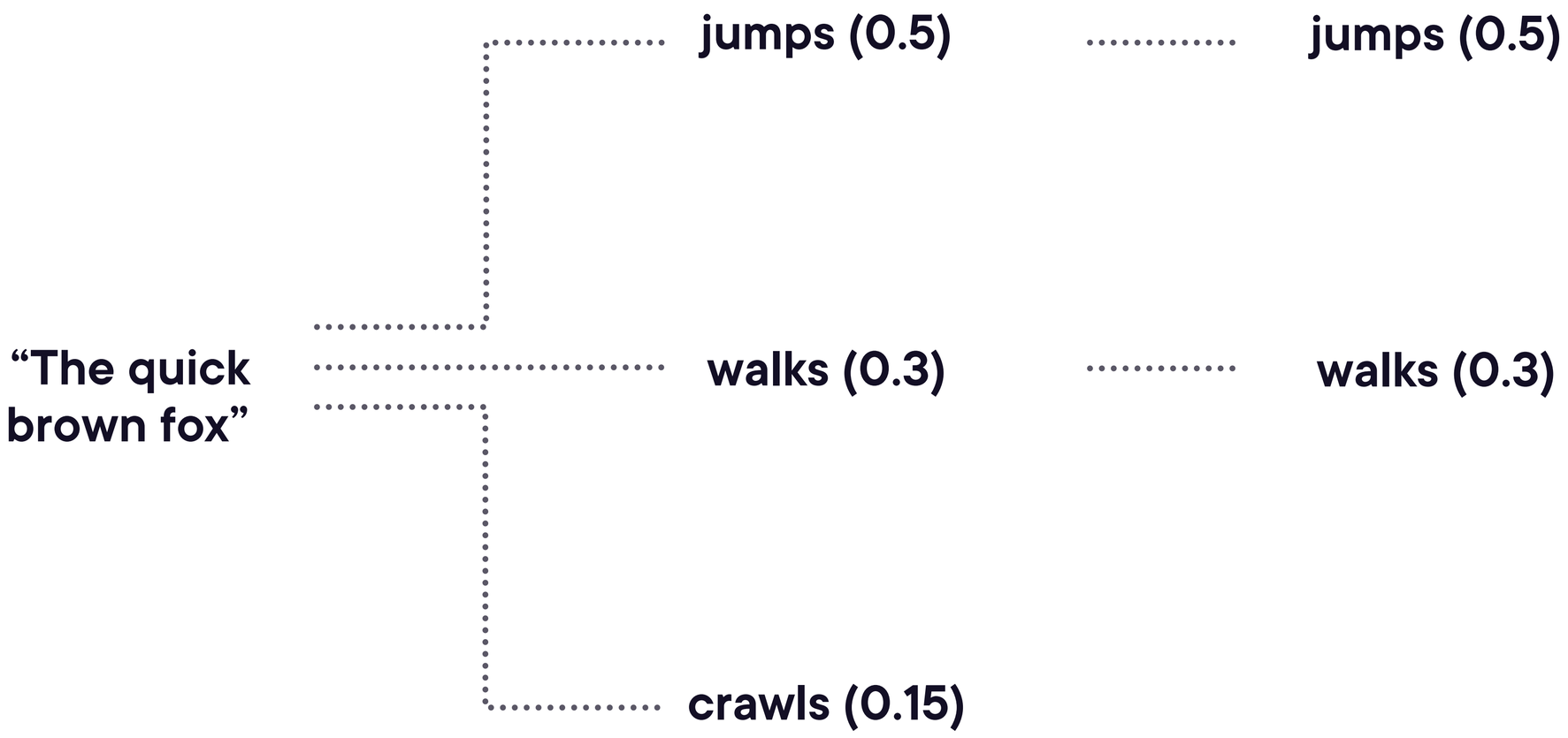
Temperature

Controls randomness of output response

Top P

Controls diversity of predicted text

Temperature and Top-P



Temperature and Top-P - Effect of High Temperature

Temperature and Top-P - Effect of Low Temperature

Demo

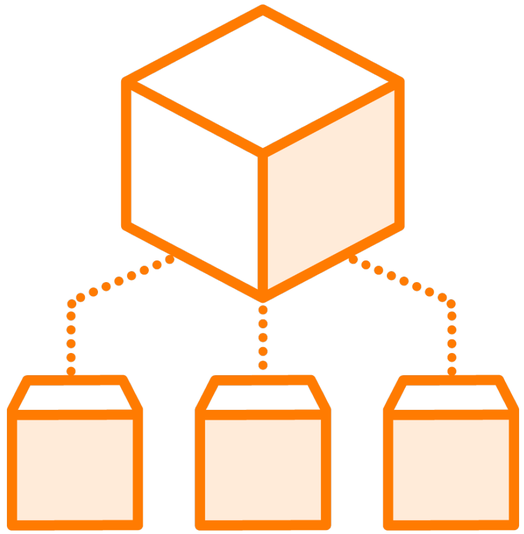
Exploring base models and parameters on Vertex

Comparing performance for

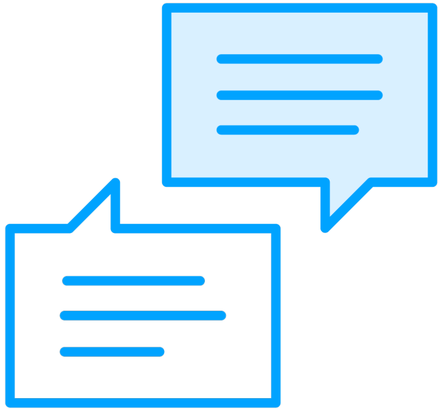
- Generating movie review descriptions
- Converting free text to JSON
- Temperature and Top-P Settings

Prompt Structure

Freeform



Structured



Chat

Demo

Exploring prompt types on Vertex

Comparing performance for

- Zero shot prompts (freeform)
- Structured prompts

Summary

Model Selection Matters

Fine-tuning Parameters

Harness the Power of Prompts